

# R3 — ML Research Track

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Member: \_\_\_\_\_

## Your role

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You own the fatembed arm (seed 0). You explain the training pipeline to the team. You build the improvise menu.

## Weekly tasks

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### Week 1 — Training pipeline explainer

- Task: Read research/tinystories/train.py fully. Document the hyperparameters: steps, target-core, seeds, arms. Explain core-matching: how arms compare at the same core budget.
- Deliverable: The 1-page explainer "How the 3 arms compare".
- Verify: R1 and R2 can explain core-matching back to you.
- Learn: Neural network basics. Write a small MLP in numpy that learns XOR.

### Week 2 — Launch fatembed seed 0

- Task: Launch the run: `uv run python -m research.tinystories.train --arm fatembed --vocab 32768 --steps 3000 --target-core 559000 --seed 0` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps.
- Learn: Embeddings and next-token prediction. Compare the ple table with the fatembed table.

### Week 3 — Collect fatembed results

- Task: Collect your two fatembed checkpoints. Record the val ppl for both seeds.
- Deliverable: The fatembed ppl numbers for both seeds.
- Verify: Your numbers sit within +/-0.05 of 11.94.
- Learn: The transformer block. Explain why a wide embedding costs memory.

### Week 4 — Improvise menu

- Task: Write the improvise menu. 3-4 directions with effort and impact. Directions: fine-tune on a new domain, serial prompting firmware, SIMD or int4 head, PLE scaling probe.
- Deliverable: IMPROVISE\_MENU.md in the club repo.
- Verify: Each direction lists the owner track and the expected effort.
- Learn: Perplexity and evaluation. Explain why fatembed sits between baseline and ple.

### Week 5 — Probe experiment

- Task: Run a small probe: change one hyperparameter (ple\_dim or learning rate) on a short run.
- Deliverable: The probe result and a 1-page finding.
- Verify: The probe compares against the same arm at the same steps.
- Learn: RoPE and KV cache. Draw how the cache grows with the context length.

### Week 6 — Analyze probe results

- Task: Analyze the probe. Write the findings page.
- Deliverable: The 1-page findings note in the club repo.
- Verify: The note states what changed and what moved.
- Learn: Fine-tuning and PTQ. Explain why a big table survives 4-bit quantization.

## Week 7 — Teach RoPE

- Task: Present RoPE and positional encoding to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: H2 can explain why position matters in a transformer.
- Learn: Prepare your presentation deeply.

## Week 8 — Next steps

- Task: Write the next-steps proposal for the club: what this project enables after month 2.
- Deliverable: NEXT\_STEPS.md in the club repo.
- Verify: Every step names the skill it builds.
- Learn: Write the "what I learned" page.

## Learning path

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1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

## Resources

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- Repo: <https://github.com/slvDev/esp32-ai>
- research/tinystories/README.md: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/README.md>
- 3Blue1Brown neural networks: [https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1\\_67000Dx\\_ZCJB-3pi](https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi)
- TinyStories paper: <https://arxiv.org/abs/2305.07759>